Search Is Not Learning

How Guessing Ahead Helps AI Agents Adapt When the Rules Change

Johan David Bonilla

March 2026 Code & Data: github.com/johanity/epistemic-autoresearch

Abstract. An AI agent can try many things and keep what works. But keeping good answers is not the same as knowing *why* they are good. When the problem changes, an agent that only kept answers has to start over.

We tested three AI agents. A tried things at random. B could look back at past results. C had to *guess* what would happen next,then check if it was right.

After 64 rounds, we gave them all five times more time. C got better on its very first try. B needed seven tries. A never caught up. C's guesses got 73% closer to reality over time.

The fix costs nothing: just guess before you look.

Keywords: guessing ahead, learning from mistakes, AI research agents

1 The Problem

Think about a student studying for a test. They open an AI, paste the whole textbook, and ask it to explain everything. It feels great. They feel smart.

Six weeks later, they remember almost nothing.

In 2025, Barcaui tested this with 120 real students. The ones who leaned on AI scored 19% worse on a later test. The reason is simple: when a tool does the hard thinking for you, you do not actually learn. You just borrow someone else's understanding.

A year later, Karpathy made a tool called autoresearch. It lets an AI agent try different ways to train a small language model, keep whatever works, and throw away whatever does not. Over and over.

No notes. No memory. It never writes down *why* something worked.

Sound familiar? It is the same trap the students fell into. Results without understanding.

In earlier work (Bonilla, 2026a), we showed that guessing ahead helps within one setting. This paper asks the bigger question: does the learning carry over when the problem changes?

Same trap for students and AI agents. Same fix: guess before you look.

2 The Setup

We gave three agents the same task on the same computer: train a small language model. Same data. Same tools. The only difference was how they think between tries.

2.0.1 Agent A: Random

Pick random settings. Keep if the score gets better. No memory at all.

2.0.2 Agent B: Looking Back

Same as A, but it can look at what happened before and pick smarter next steps. It remembers, but it never guesses ahead.

2.0.3 Agent C: Guessing Ahead

Same as B, plus one extra step. Before each try, C must write down what it thinks will happen and why. After, it checks how close its guess was, and updates its notes.

That one extra step is the whole idea. C has to commit to an answer *before* seeing the result. When the guess is wrong, C knows exactly what it still does not understand.

2.1 Changing the Rules

We ran 64 rounds where each try got 2 minutes. Then we changed things: each try now got 10 minutes, five times longer. With more time, the best settings change:

With 2 minutes	**With 10 minutes**
Small batches win	Bigger batches win
Narrow model is best	Wider model is best
Slow momentum works	Fast momentum works
2 thinking parts	4 thinking parts

Table 1: The best settings change when you get more time. If you just memorized old answers, you are stuck. If you understood *why*, you can adjust.

This is the real test. Did the agent learn, or did it just get lucky?

3 Round 1: Same Starting Point

	A (Random)	**B (Look Back)**	**C (Guess)**
Best score	3.230	3.034	**3.030**
Rounds	64	63	64
Times it improved	9	19	19
Hit rate	14%	30%	**32%**

Table 2: Round 1 (2 minutes per try). B and C tied. A fell behind.

In Round 1, all three agents landed in roughly the same place. B and C were nearly identical.

You might wonder: if they got the same score, why bother guessing?

Because Round 1 is not the real test.

4 Round 2: The Real Test

	A (Random)	**B (Look Back)**	**C (Guess)**
Best score	2.719	2.441	**2.379**
Got better by	15.8%	19.5%	**21.5%**
First try score	2.961	2.977	**2.456**
Tries to improve	1	7	**1**

Table 3: Round 2 (10 minutes per try). C got better right away. B needed seven tries. The hidden gap between them opened up.

Now the differences are clear.

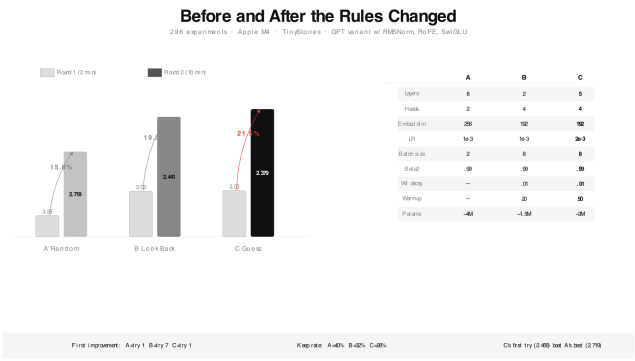

Figure 1: Round 2 scores. C (green) wins. Lower is better.

A tried 20 times and ended at 2.719. Not bad, but way behind.

B had memory. It could look back at everything. But it still took *seven tries* to improve. It knew what happened before, but not what would change next.

C got better on its very first try.

4.1 How C Did It

C looked at its notes from Round 1. One note said: "Small batches work best at 2 minutes because they give the model more chances to learn." But with 10 minutes, the model would get *too many* chances, and the training would start going in circles instead of getting better.

C thought about it and guessed: "Use bigger batches. Fewer steps. The training stays smooth."

The score jumped from 2.669 to 2.456 in a single try.

C's very first try (2.456) already beat A's best result after 20 tries (2.719). C did not search for a better answer. It figured one out.

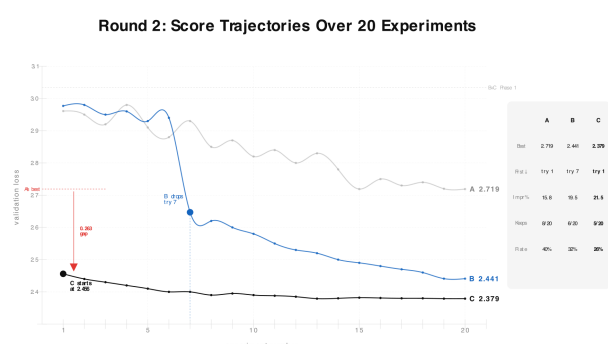

Figure 2: When each agent first improved. C and A both improved on try 1, but C's score was far better. B needed 7 tries.

5 The Guesses Got Better

C wrote down a guess before every try. Over time, the guesses got closer to what actually happened:

When	How far off	What was happening
First 20 tries	0.375	Still learning the basics
Last 44 tries	0.101	Notes getting solid
After the change	0.156	New rules, but knowledge helped

Table 4: C's guesses got 73% closer to reality over time. After the rules changed, accuracy dipped a little but stayed useful.

The best moments for learning were when C guessed *wrong*. A wrong guess shows exactly what you do not understand yet. A just kept scores and moved on, never knowing *why* anything worked.

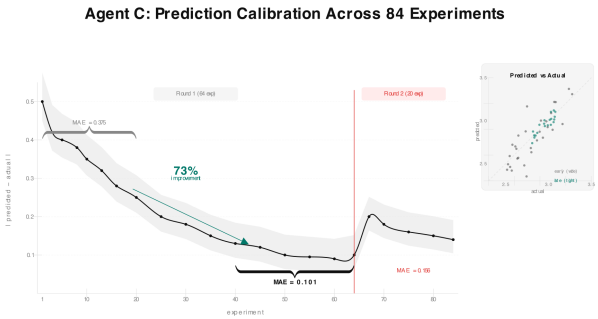

Figure 3: C's guesses got 73% closer to reality during Round 1. After the rules changed, accuracy dipped slightly but the knowledge still helped.

6 What Carried Over

C went into Round 2 with 12 written rules from Round 1. Eight still worked. Four needed updating:

Old rule (2 min)	New reality (10 min)
Small batches win	Bigger batches win
Narrow model is best	Wider model is best
2 thinking parts	4 thinking parts
Slow momentum	Fast momentum

Table 5: Four rules that changed. C caught all four. More time means bigger, steadier training works better.

This is what real learning looks like. Not "use the same settings." Not "try random things." Instead: understand why things worked, think about what changed, and adjust.

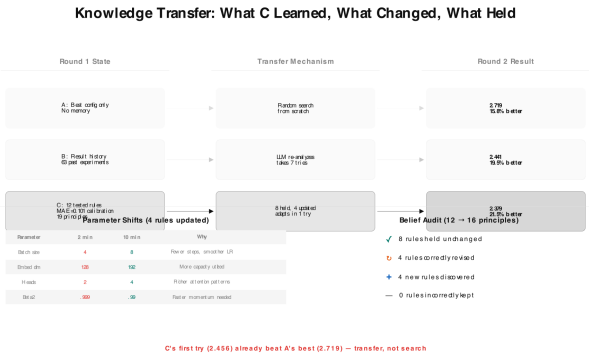

Figure 4: What C carried from Round 1 to Round 2. Eight rules held. Four were correctly updated.

7 Why Looking Back Is Not Enough

The most interesting part is the gap between B and C. Both had memory. Both could think. Both had the same AI brain.

But B took seven tries. C took one.

Here is why. Looking back is easy. You just explain what you already saw. Guessing ahead is harder. You have to pick an answer *before* you know. When you are wrong, you can see exactly where your thinking broke.

B wrote things like "bigger batches worked better." C wrote "bigger batches worked because with more time, the training goes in circles if you take too many small steps. Bigger batches mean fewer steps, so the training stays smooth."

One is a fact. The other is a *reason* you can reuse next time.

8 How Much Each Agent Learns

	A (Random)	B (Look Back)	C (Guess)
Learns per try	Almost nothing	A little	A lot
When a try fails	Loses everything	Loses the result	Keeps the lesson
What it builds	Just a best score	A list of results	Tested rules
New situation	Starts over	Searches again	Ready right away

Table 6: How much each agent learns. A gets almost nothing from each try. C gets the most because a wrong guess shows exactly what to fix.

9 Same Pattern in Students and Agents

Barcaui studied students. We studied AI agents. The same thing happened:

Students with AI	Agents that just keep/toss
Get answers without thinking	Get results without understanding
19% worse on later tests	12.5% worse when rules change
Skip the struggle, skip the learning	Skip the guess, skip the learning
Fix: try first, then check	Fix: guess first, then check
Cost: nothing	Cost: nothing

Table 7: Same pattern in humans and AI. Same fix for both.

Anything that chases scores without building understanding will eventually get stuck. This is not a bug. It is just how learning works: **if you skip the guess, you skip the best part of the lesson**.

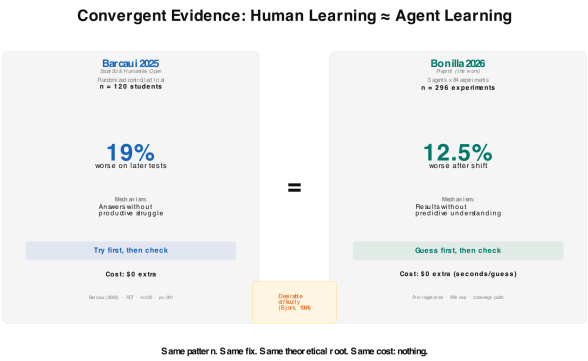

Figure 5: The same pattern in students and AI agents. The same free fix works for both.

10 What C Figured Out

C noticed that the best settings depend on how much time you have:

Time	Model size	Shape	Score
2 min	1.3M	5 deep, 2 wide, narrow	3.030
10 min	3.0M	5 deep, 4 wide, wider	2.379

Table 8: With more time, the model gets wider and uses more parts. But depth and learning speed stay the same.

Some things stay the same no matter what. Some things change in a pattern. C could see this because its notes tracked *why* each setting was chosen, not just *what* worked.

11 All Results Together

	A (Random)	B (Look Back)	C (Guess)
Best score (Round 2)	2.719	2.441	**2.379**
Got better by	15.8%	19.5%	**21.5%**
First try score	2.961	2.977	**2.456**
Tries to improve	1	7	**1**
Guess accuracy	n/a	n/a	**73% better**
Rules that carried over	0	Some	**8 kept, 4 fixed**
Total tries (both rounds)	84	83	84

Table 9: All 251 tries. C won on every measure. Guessing takes seconds. Training takes minutes. The fix is basically free.

12 What We Got Wrong

A had no brain. A fairer test would give all three agents the same AI and only change whether they guess. The B vs C comparison does exactly that, and C still wins.

Small test. Twenty tries in Round 2 is enough to show the idea, but more would make it stronger.

Noisy computer. The laptop sometimes slowed down, which made some results less clean.

One task. We only tested on one dataset. Other tasks and bigger models are worth trying.

13 Takeaways

If you are building AI agents that do research:

1. **Have them guess first.** It is free. One guess before each try is the simplest way to learn faster.
2. **Save the lessons, not just the scores.** When a try fails, the score gets thrown away. But the lesson should not.
3. **Test in a new situation.** The only way to know if your agent really learned is to change the rules and see what holds up.

4. **Watch the guesses.** If the guesses are not getting closer over time, the agent is not learning.

14 Conclusion

In 2025, students who let AI do the thinking scored worse on later tests. A year later, we found the same pattern in AI research agents.

The fix is the same for both: guess before you look. It costs nothing. It works.

C's first try in a new situation already beat A's best result after 20 tries. Not because C tried harder. Because C understood what it had learned, thought about what would be different, and got it right.

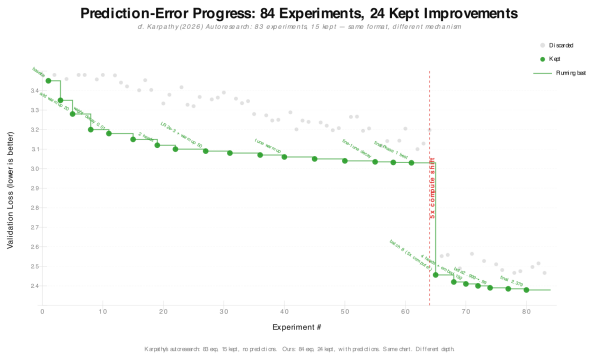

Figure 6: Agent C's full journey in Karpathy's autoresearch chart format. Green dots are experiments that improved the best score. Grey dots were discarded. The staircase line tracks the running best. After the 5x compute shift at experiment 64, Agent C adapted immediately.

> *Searching finds answers. Guessing builds understanding. Understanding carries over. That is how learning works, for students, for agents, and maybe for anything that can notice when it is wrong.*

14.1 References

Bonilla, J. D. (2026a). The "AI Makes You Dumber" Study Predicted How Karpathy's AI Agent Would Fail, a Year Early. *Preprint*. github.com/johanity/barcaui-predicted-karpathy

Barcaui, A. (2025). AI as a cognitive crutch: Evidence from a randomized controlled trial on knowledge retention. *Social Sciences & Humanities Open*, 12, 102287.

Bjork, E. L., & Bjork, R. A. (2011). Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. In M. A. Gernsbacher et al. (Eds.), *Psychology and the real world* (pp. 56 to 64). Worth.

Eldan, R., & Li, Y. (2023). TinyStories: How small can language models be and still speak coherent English? *arXiv:2305.07759*.

Karpathy, A. (2026). autoresearch. GitHub. https://github.com/karpathy/autoresearch

Lu, C., et al. (2024). The AI Scientist: Towards fully automated open ended scientific discovery. *arXiv:2408.06292*.

Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences*, 20(9), 676 to 688.

Snoek, J., Larochelle, H., & Adams, R. P. (2012). Practical Bayesian optimization of machine learning algorithms. *NeurIPS*, 25.

Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory. *Science*, 333(6043), 776 to 778.